Sales Analytics Pipeline

Phase 2 — Data Modeling (ERD)

Entity relationships and key reference across Bronze, Silver, Gold, and Export layers

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Revision: v3 — CUSTOM_ACTIVITIES is replaced by CUSTOM_ACTIVITY_TYPES + CUSTOM_ACTIVITY_FIELDS after reviewing the project's explanation video and GoodNotes transcript; see Sections 4.1, 4.2, 7

Status: Draft — pending SME review and sign-off before Phase 3

Builds on: Phase 1 — Architecture Design (v3)

1. Purpose & Scope

This document completes the Phase 2 deliverable: a full entity relationship model for the Sales Analytics Pipeline, covering Bronze, Silver, Gold, and Export. It builds directly on the Phase 1 architecture (table/view names and layer design) and the provided Snowflake Data Architecture Model, and incorporates the leads_raw decision confirmed with the SME: leads_raw is ingested to Bronze, with Silver/Gold modeling deferred until a report requires it.

Two diagrams are provided: a formal entity relationship diagram for the Silver layer (primary keys, foreign keys, and cardinality — the layer where real relational structure exists), and a source-to-report lineage diagram for Gold and Export (which are business-logic views and materialized aggregates, not physically keyed tables). Both are followed by a written table/key reference so every relationship shown visually is also stated explicitly.

2. How to Read This ERD

- PK (primary key) — the column, or combination of columns, that uniquely identifies a row in that table.
- FK (foreign key) — a column whose value should correspond to a PK in another table. An arrow label of "(deferred)" means the source column exists but the target entity is not yet modeled in Silver.
- Cardinality (1, N, 0..1) — read from the entity nearest the label toward the other end, e.g. CLOSE_CRM_USERS_PROCESSED "1" to LEAD_ACTIVITIES_PROCESSED "N" means one user is linked to many activities.
- Bronze tables and Gold/Export objects are not part of the formal ERD because they are not relationally keyed: Bronze is an unstructured JSON landing zone, and Gold/Export are views and aggregates derived by filtering and joining Silver. Both are still documented with grain and lineage below.

3. Bronze Layer Entities

Bronze tables carry no primary key — every row is a raw JSON payload plus a load timestamp, and the same logical record can appear many times across daily batches. Grain is per-file-record-per-load, not per-business-entity.

Bronze Table	Grain	Feeds
LEAD_ACTIVITIES_RAW	1 row per activity record per daily load	LEAD_ACTIVITIES_PROCESSED_TRANSIENT
CUSTOM_ACTIVITIES_RAW	1 row per activity-type-definition record per daily load	CUSTOM_ACTIVITIES_TRANSIENT -> CUSTOM_ACTIVITY_TYPES, CUSTOM_ACTIVITY_FIELDS
CLOSE_CRM_USERS_RAW	1 row per user record per daily load	CLOSE_CRM_USERS_PROCESSED
LEADS_RAW (added, Phase 1 Sec. 4.1)	1 row per lead record per daily load	None yet — landed for future use, not modeled further in this phase

4. Silver Layer — Entities & Relationships

The Silver layer is where the pipeline first has real relational structure: deduplicated, typed tables with primary keys, joined by shared business identifiers.

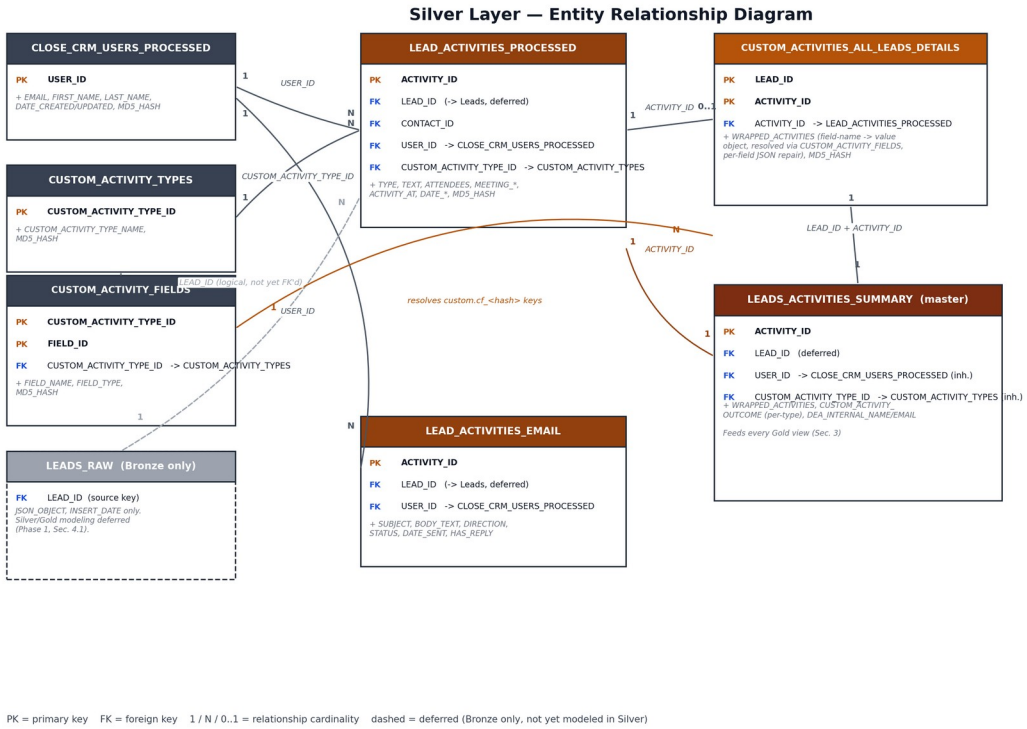


Figure 1. Silver layer entity relationship diagram (v3 — two-level custom activity dictionary).

Revised from v2: CUSTOM_ACTIVITIES (a flat outcome dictionary) is replaced by CUSTOM_ACTIVITY_TYPES and CUSTOM_ACTIVITY_FIELDS. This follows from reviewing the project's explanation video and GoodNotes transcript, which showed that custom field values on an activity are keyed by an opaque custom.cf_<hash> ID, and that ID only resolves to a name via a per-activity-type field list in custom_activites_raw — a two-level dictionary, not the one-level outcome lookup originally modeled.

4.1 Entity Reference

Entity	Primary Key	Foreign Keys	Grain
CLOSE_CRM_USERS_PROCESSED	USER_ID	—	1 row per CRM user
CUSTOM_ACTIVITY_TYPES	CUSTOM_ACTIVITY_TYPE_ID	—	1 row per custom activity type (Triage Call, Strategy Call, ...)
CUSTOM_ACTIVITY_FIELDS	CUSTOM_ACTIVITY_TYPE_ID + FIELD_ID	CUSTOM_ACTIVITY_TYPE_ID -> CUSTOM_ACTIVITY_TYPES	1 row per field defined on a custom activity type
LEAD_ACTIVITIES_PROCESSED	ACTIVITY_ID	USER_ID -> CLOSE_CRM_USERS_PROCESSED; CUSTOM_ACTIVITY_TYPE_ID -> CUSTOM_ACTIVITY_TYPES; LEAD_ID, CONTACT_ID (deferred)	1 row per deduplicated activity
LEAD_ACTIVITIES_EMAIL	ACTIVITY_ID	USER_ID ->	1 row per email/SMS

		CLOSE_CRM_USERS_PROCESSED; LEAD_ID (deferred)	activity
CUSTOM_ACTIVITIES_ALL_LEADS_DETAILS	LEAD_ID + ACTIVITY_ID	ACTIVITY_ID -> LEAD_ACTIVITIES_PROCESSED; resolved per-field via CUSTOM_ACTIVITY_FIELDS	1 row per lead's custom activity
LEADS_ACTIVITIES_SUMMARY	ACTIVITY_ID	USER_ID -> CLOSE_CRM_USERS_PROCESSED; CUSTOM_ACTIVITY_TYPE_ID -> CUSTOM_ACTIVITY_TYPES; LEAD_ID (deferred)	1 row per activity, fully resolved (master table)

4.2 Relationship Reference

From	To	Join Key	Cardinality	Notes
CLOSE_CRM_USERS_PROCESSED	LEAD_ACTIVITIES_PROCESSED	USER_ID	1 : N	One user performs many activities
CLOSE_CRM_USERS_PROCESSED	LEAD_ACTIVITIES_EMAIL	USER_ID	1 : N	
CUSTOM_ACTIVITY_TYPES	LEAD_ACTIVITIES_PROCESSED	CUSTOM_ACTIVITY_TYPE_ID	1 : N	Populated only where TYPE = CustomActivity
CUSTOM_ACTIVITY_TYPES	CUSTOM_ACTIVITY_FIELDS	CUSTOM_ACTIVITY_TYPE_ID	1 : N	Each type defines its own field list
CUSTOM_ACTIVITY_FIELDS	CUSTOM_ACTIVITIES_ALL_LEADS_DETAILS	CUSTOM_ACTIVITY_TYPE_ID + FIELD_ID	1 : N	Logical, not a stored FK — used to resolve custom.cf_<hash> keys into WRAPPED_ACTIVITIES
LEAD_ACTIVITIES_PROCESSED	CUSTOM_ACTIVITIES_ALL_LEADS_DETAILS	ACTIVITY_ID	1 : 0..1	Only custom activities carry wrapped fields
LEAD_ACTIVITIES_PROCESSED	LEADS_ACTIVITIES_SUMMARY	ACTIVITY_ID	1 : 1	Enriched version of the same activity row
CUSTOM_ACTIVITIES_ALL_LEADS_DETAILS	LEADS_ACTIVITIES_SUMMARY	LEAD_ID + ACTIVITY_ID	1 : 1	
LEADS_RAW (Bronze)	LEAD_ACTIVITIES_PROCESSED / LEADS_ACTIVITIES_SUMMARY	LEAD_ID	1 : N (logical)	Not an enforced constraint — deferred per Phase 1 decision

5. Gold Layer — Views & Lineage

Gold views are not physically keyed tables; each is a filtered, joined read of LEADS_ACTIVITIES_SUMMARY (plus CUSTOM_ACTIVITY_TYPES for activity-type text and CLOSE_CRM_USERS_PROCESSED for setter/closer identity), scoped to one stage of the funnel. The diagram below shows lineage — what each view is built from — rather than primary/foreign key pairs. Setter/Closer on ALL_STRATEGIES_DETAILS and SALES_DETAILS now resolve primarily from that activity's own Setter/Closer custom fields (via CLOSE_CRM_USERS_PROCESSED), not only by looking back at the booking view (Phase 3 deviation, see runbook Section 8).

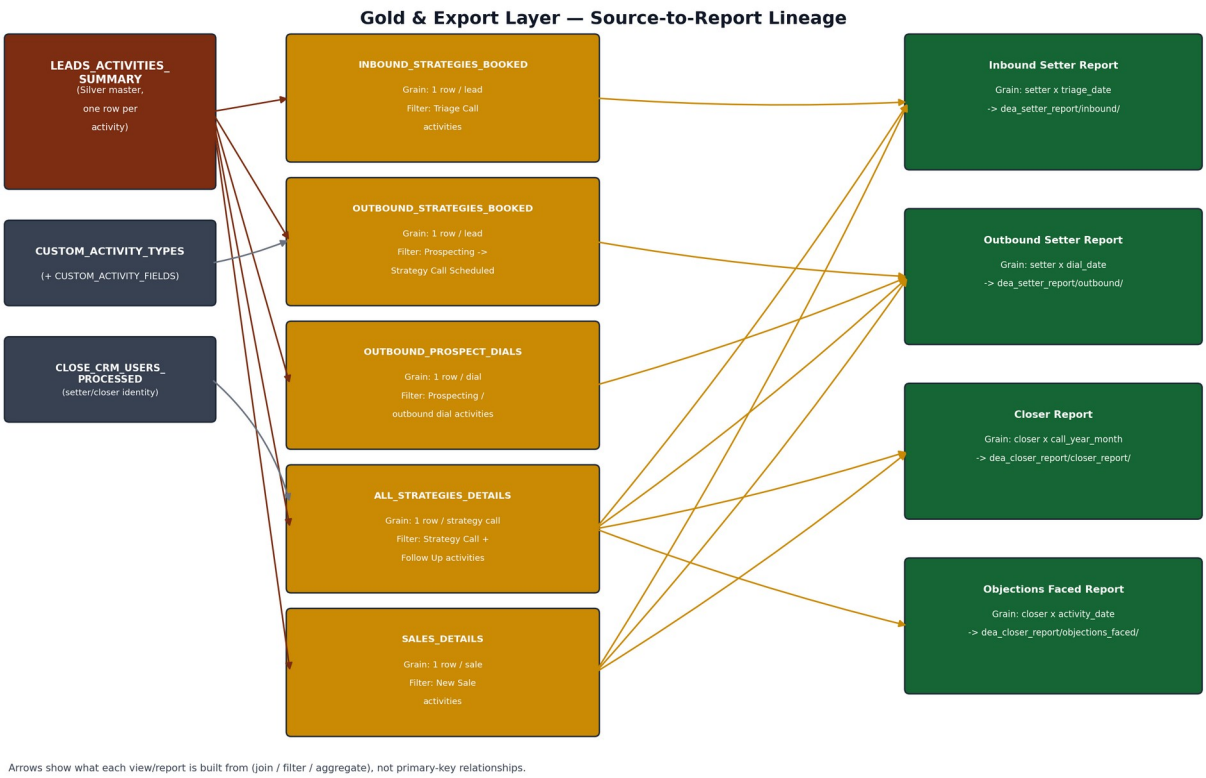


Figure 2. Gold and Export layer source-to-report lineage.

5.1 Gold View Reference

View	Grain	Filter / Business Rule
INBOUND_STRATEGIES_BOOKED	1 row per lead (inbound path)	Triage Call activities; STRATEGY_CALL_BOOKED = outcome '1. Strategy Call Scheduled'
OUTBOUND_STRATEGIES_BOOKED	1 row per lead (outbound path)	Prospecting/outbound call activities; outcome '2. Strategy Call Scheduled'
OUTBOUND_PROSPECT_DIALS	1 row per dial activity	All prospecting/outbound dial activities regardless of outcome (top-of-funnel volume)
ALL_STRATEGIES_DETAILS	1 row per strategy call activity	Strategy Call + Strategy Call Follow Up activities; excludes no-show/cancel from Show metrics; OFFER_PRESENTED derived
SALES_DETAILS	1 row per sale activity	New Sale activities where a prior attended Strategy Call exists for the same lead

6. Export Layer — Report Reference

Report	Grain	Built From	Target Stage
Inbound Setter Report	setter x triage_date	INBOUND_STRATEGIES_BOOKED, ALL_STRATEGIES_DETAILS, SALES_DETAILS	dea_setter_report/inbound/
Outbound Setter Report	setter x dial_date	OUTBOUND_PROSPECT_DIALS, OUTBOUND_STRATEGIES_BOOKED, ALL_STRATEGIES_DETAILS, SALES_DETAILS	dea_setter_report/outbound/
Closer Report	closer x call_year_month	ALL_STRATEGIES_DETAILS, SALES_DETAILS	dea_closer_report/closer_report/
Objections Faced Report	closer x activity_date	ALL_STRATEGIES_DETAILS (objection fields)	dea_closer_report/objections_faced/

7. Cross-Layer Key Reference

A single reference for the four identifiers used throughout the model:

Key	Originates In	Used By	Notes
LEAD_ID	leads_raw / lead_activities payload	LEAD_ACTIVITIES_PROCESSED, LEAD_ACTIVITIES_EMAIL, CUSTOM_ACTIVITIES_ALL_LEADS_DETAILS, LEADS_ACTIVITIES_SUMMARY, all Gold views	No Silver LEADS entity yet; treated as an attribute, not an enforced FK, until leads_raw is modeled (Sec. 8)
ACTIVITY_ID	lead_activities_raw payload	LEAD_ACTIVITIES_PROCESSED (PK), CUSTOM_ACTIVITIES_ALL_LEADS_DETAILS, LEADS_ACTIVITIES_SUMMARY (PK)	Deduplicated with LEAD_ID via MD5_HASH + keep-latest ACTIVITY_AT (Phase 1, Sec. 5.5)
USER_ID	close_crm_users_raw payload	CLOSE_CRM_USERS_PROCESSED (PK), LEAD_ACTIVITIES_PROCESSED, LEAD_ACTIVITIES_EMAIL, LEADS_ACTIVITIES_SUMMARY	Setter vs. closer role is resolved per Gold view, not stored as a static attribute
CUSTOM_ACTIVITY_TYPE_ID	custom_activites_raw payload (top-level id)	CUSTOM_ACTIVITY_TYPES (PK), CUSTOM_ACTIVITY_FIELDS (PK part), LEAD_ACTIVITIES_PROCESSED, LEADS_ACTIVITIES_SUMMARY	Identifies the activity type (Triage Call, Strategy Call, …); no longer paired with an OUTCOME_ID
FIELD_ID (custom.cf_<hash>)	custom_activites_raw payload (nested fields[].id)	CUSTOM_ACTIVITY_FIELDS (PK part); matched against each custom.cf_<hash> key on an activity record	Opaque, Close-generated; only meaningful when resolved per CUSTOM_ACTIVITY_TYPE_ID (Sec. 4.1)

8. Open Items for SME Review

- Confirm whether ACTIVITY_ID is globally unique or only unique within a LEAD_ID. The Phase 1 dedup logic keys on LEAD_ID + ACTIVITY_ID, while LEAD_ACTIVITIES_PROCESSED and LEADS_ACTIVITIES_SUMMARY are modeled here with ACTIVITY_ID alone as PK — these should agree before Phase 3 implements the MERGE statements.
- CONTACT_ID appears on LEAD_ACTIVITIES_PROCESSED but there is no CONTACTS entity in the current model. Confirm whether contact-level detail is needed for any of the four reports or can remain unmodeled, consistent with the leads_raw deferral.
- Resolved (v3): CUSTOM_ACTIVITIES is replaced by CUSTOM_ACTIVITY_TYPES + CUSTOM_ACTIVITY_FIELDS, confirmed against the explanation video/transcript. Still open: confirm the field-ID -> field-name mapping is stable over time per activity type (i.e. Close doesn't reuse a cf_<hash> for a different field on the same type later), since CUSTOM_ACTIVITY_FIELDS is upserted by (type, field) key.
- Confirm LEAD_ACTIVITIES_PROCESSED to LEADS_ACTIVITIES_SUMMARY is strictly 1:1 (every processed activity gets a summary row) rather than filtered to a subset of activity types.
- Once leads_raw is modeled beyond Bronze, revisit LEAD_ID everywhere in this ERD as a proper enforced foreign key rather than a shared attribute.

9. Next Steps

- Phase 3 — Build the pipeline: Bronze COPY INTO, Silver repair/FLATTEN/MERGE (including the JSON repair UDF and MD5_HASH dedup), Gold view SQL implementing the funnel filters above, and Export unload jobs, orchestrated with Snowflake Tasks and Streams. Run daily for 7 days to simulate production before handoff, per the original requirements.